

Introduction to Retrieval-Augmented Generation (RAG)

What is RAG?

Retrieval-Augmented Generation (RAG) is a technique that combines a retrieval system with a large language model. Instead of relying only on the model's training data, RAG fetches relevant context from an external knowledge base and provides it to the model at inference time.

Why use RAG?

RAG reduces hallucination, allows the model to use up-to-date information, and makes it possible to cite sources. It is much cheaper than fine-tuning a model on private data and the knowledge base can be updated without retraining.

Core components

A RAG pipeline has four pieces: a document loader, a text splitter that creates chunks, an embedding model that turns each chunk into a vector, and a vector store such as FAISS that supports nearest-neighbor search.

Typical workflow

At query time the user question is embedded, the top K most similar chunks are retrieved from the vector store, and those chunks are passed as context inside the prompt to the language model. The model then generates an answer grounded in that context.

Common pitfalls

Chunk size matters: chunks that are too small lose context, chunks that are too large dilute the signal. A typical starting point is 500 characters with 50 characters of overlap. Quality of embeddings also directly affects retrieval quality.